

# **Phylogenetic Analysis Method**

**Shatakshi Kulkarni**

# Introduction

- ❑ Phylogenetic tree analysis is a method used to study the evolutionary relationships between different species or organisms.
- ❑ 3 methods for constructing phylogenetic trees:
  - **UPGMA/Unweighted Pair Group Method with Arithmetic Mean**
  - **NJ/Neighbor Joining method**
  - **Maximum Likelihood method**

# UPGMA/Unweighted Pair Group Method with Arithmetic Mean

## ❑ Overview:

1. UPGMA is a simple, distance-based method for constructing phylogenetic trees.
2. It assumes a constant rate of evolution (molecular clock assumption), meaning that **all lineages evolve at the same rate**.
3. This method works by clustering sequences or species with the smallest pairwise distances first and progressively grouping clusters based on average distances.

## ❑ Steps:

1. **Calculate the pairwise distance matrix for all sequences. This matrix is often based on differences between sequences (like mismatches or substitutions).**
2. **Identify the smallest distance** in the matrix and merge the corresponding sequences or clusters.
3. **Calculate the average distance** of the new cluster to all other sequences or clusters.
4. **Update the matrix** with the new cluster, and repeat the process until all sequences are merged into a single cluster.
5. **Build the tree** based on the order of merges and the distances.

❑ **Advantage:** Simple and fast, suitable for small datasets

❑ **Disadvantage:** Assumes a constant rate of evolution, which is often unrealistic

❑ **Use Case E.g.:** In early studies of species or gene relationships where evolutionary rates are assumed to be constant, or for smaller datasets where simplicity is prioritized over accuracy.

## UPGMA E.g.,

### Phylogenetic Tree:

seq 1    A T C G A T C G  
seq 2    G T A G A C G A  
seq 3    A C C G T A C G  
seq 4    T C A G T C A G  
seq 5    G C C T A C A G

I.) fill matrix by calculating mismatches

|   | A | B | C   | D | E |
|---|---|---|-----|---|---|
| A | 0 | 5 | (3) | 6 | 5 |
| B |   | 0 | 7   | 5 | 6 |
| C |   |   | 0   | 4 | 5 |
| D |   |   |     | 0 | 4 |
| E |   |   |     |   | 0 |

II.) start with smallest distance in matrix  
(excluding 0) → here it is AC (3)

III.) Merge the closest pair (A & C) & create a  
new node (A-C)

AC considered as one group with new average  
distance.

# UPGMA E.g.,

Phylogenetic Tree:

seq 1 ATCGATCG  
 seq 2 GTAGACGA  
 seq 3 ACCGTACG  
 seq 4 TCAGTLAG  
 seq 5 GCCTACAG

I.) fill matrix by calculating mismatches

|   | A | B | C   | D | E |
|---|---|---|-----|---|---|
| A | 0 | 5 | (3) | 6 | 5 |
| B |   | 0 | 7   | 5 | 6 |
| C |   |   | 0   | 4 | 5 |
| D |   |   |     | 0 | 4 |
| E |   |   |     |   | 0 |

II.) start with smallest distance in matrix (excluding 0) → here it is AC (3)

III.) Merge the closest pair (A & C) & create a new node (A-C)  
 AC considered as one group with new average distance.

Date:

|    | AC | B | D | E |
|----|----|---|---|---|
| AC | 0  | 6 | 5 | 5 |
| B  |    | 0 | 5 | 6 |
| D  |    |   | 0 | 4 |
| E  |    |   |   | 0 |

AC with B =  $\frac{A \text{ with } B + C \text{ with } B}{2}$   
 $= \frac{5 + 7}{2} = 6$

AC with D =  $\frac{A \text{ with } D + C \text{ with } D}{2}$   
 $= \frac{6 + 4}{2} = 5$

AC with E =  $\frac{A \text{ with } E + C \text{ with } E}{2}$   
 $= \frac{5 + 5}{2} = 5$

IV. Find the next closest pair (smallest value)  
 Here, D & E with 4 value

## UPGMA E.g.,

V) Merge the closest pair (D & E)

Date:

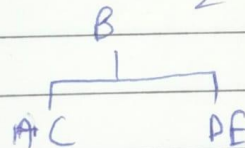
& create a new pair DE:

|     | A C | B | DE  |
|-----|-----|---|-----|
| A C | 0   | 6 | 5   |
| B   |     | 0 | 5.5 |
| DE  |     |   | 0   |

AC with B = from previous table

$$\begin{aligned} \text{AC with DE} &= \frac{\text{AC with D} + \text{AC with E}}{2} \\ &= \frac{5+5}{2} = 5 \end{aligned}$$

$$\begin{aligned} \text{B with DE} &= \frac{\text{B with D} + \text{B with E}}{2} \\ &= \frac{5+6}{2} = \frac{11}{2} = 5.5 \end{aligned}$$



# NJ/Neighbor Joining Method

## ❑ Overview:

1. The Neighbor-Joining method is a distance-based algorithm for constructing phylogenetic trees without assuming a constant rate of evolution.
2. Instead, it tries to minimize the total branch length of the tree, making it more flexible and applicable to real-world data where evolutionary rates vary. This best explains the observed genetic distances with the fewest evolutionary changes.

## ❑ Steps:

1. Compute the pairwise distance matrix for all sequences and create matrix that accounts for total divergence of each sequence.
2. Identify the pair of sequences or clusters that minimizes the total branch length and join them as a new cluster.
3. Calculate the distance of the new cluster to all remaining sequences or clusters.
4. Update the matrix with the new cluster and repeat until a single cluster remains.
5. Build the tree by arranging clusters based on the order of merges and calculated distances.

❑ **Advantage:** Does not assume a constant rate of evolution, so it's more accurate for real-world data.

❑ **Disadvantage:** Can sometimes produce less accurate trees if the pairwise distances are not sufficiently informative.

❑ **Use Case E.g.:** studies of genetic sequences across diverse species.

# Maximum Likelihood

## ❑ Overview:

1. Maximum Likelihood is a character-based method that finds the tree structure most likely to have produced the observed data based on a specified model of sequence evolution.
2. Unlike UPGMA and NJ, which are distance-based, ML works directly with sequence data and uses statistical models to estimate evolutionary relationships.

## ❑ Steps:

1. Define an evolutionary model (The evolutionary model acts as a statistical tool to interpret patterns in the sequence data, guiding the tree-building process by explaining how likely each mutation or arrangement of sequences is, given an assumed evolutionary process)
2. Generate possible tree topologies
3. Calculate the likelihood of each tree given the observed data, and select the tree with the highest likelihood as the best estimate of the evolutionary relationships.

❑ **Advantage:** Can handle complex models of evolution and sequence data

❑ **Disadvantage:** Computationally intensive, especially for large datasets, because it needs to evaluate multiple tree topologies & models.

❑ **Use Case E.g.:** Maximum Likelihood is often used in evolutionary studies where precise evolutionary relationships are needed, such as reconstructing ancestral states, estimating divergence times, or analyzing highly variable sequences like viral genomes.



# **Alignment Analysis tools**



# CINEMA (Comprehensive INteractive Multiple Alignment)

- ❑ CINEMA is a multiple sequence alignment analysis tool that provides an interactive and user-friendly interface for visualizing and editing sequence alignments.
- ❑ It's designed to allow users to explore, refine, and **understand the relationships between sequences** by aligning them more accurately.
- ❑ CINEMA supports color-coding for highlighting specific features like conserved residues, facilitating a clearer interpretation of evolutionary relationships and **functional similarities across sequences**.
- ❑ It is often used in research requiring precise alignment adjustments for detailed comparative studies in molecular biology and evolutionary analysis.

# MaxAlign

- ❑ MaxAlign is a tool for improving the quality of multiple sequence alignments by focusing on conserved, reliable regions across the sequences.
- ❑ Unlike traditional alignment tools, MaxAlign optimizes the alignment by **eliminating sequences that introduce noise, maximizing the conservation across the remaining sequences.**
- ❑ This approach enhances the reliability of comparative analyses, phylogenetic studies, and protein structure prediction.
- ❑ It is particularly useful when working with divergent sequences or when high alignment accuracy is required, as it helps **focus the alignment on the most informative sequences.**